EESC GU4220 Glaciology: Final Project Proposal (Anna Chen, yc4406)

Project Title: *Quantifying the Contribution of Elevation to $\delta^{18}O$ Variability During the Penultimate Deglaciation and the Last Interglacial Using 6 Antarctic Ice Cores*

Background and Objective

The $\delta^{18}O$ composition preserved in Antarctic ice cores is a fundamental paleothermometer, reflecting the isotopic fractionation that occurs as moist air masses cool and precipitate from low latitudes to the poles. However, $\delta^{18}O$ is not solely governed by global temperature or ice-volume changes. Variations in ice-sheet elevation alter the condensation altitude and local temperature of snowfall, introducing a fractionation effect due to elevation that can amplify or dampen the apparent climatic signal.

During the penultimate deglaciation (140-129 ka) and the last interglacial (129-115 ka), when the Antarctic Ice Sheet thinned substantially, surface elevation changes may have contributed a nontrivial component to $\delta^{18}O$ anomalies traditionally attributed to global warming. This project aims to isolate and quantify this "elevation effect" in order to better constrain the temperature interpretation of isotopic records from Antarctic ice cores.

Data Source

a) **Ice core data:** $\delta^{18}O$ in H_2O time series from 1) Dome Fuji (3810m), 2) EPICA Dome C (3233m), 3) EPICA Dronning Maud Land (2891m), 4) Talos Dome (2315m), 5) Taylor Dome (2365m), 6) Vostok (3488m)
   - Source: Masson-Delmotte, V et al. (2011): A comparison of the present and last interglacial periods in six Antarctic ice cores. Climate of the Past, 7, 397-423, https://doi.org/10.5194/cp-7-397-2011.
     PANGAEA Link: https://doi.pangaea.de/10.1594/PANGAEA.785228.

b) **Surface and bed topography (modern):** *P6_data.npz* from Practical 6

Methods

The project will use a Glen's flow–based 1-D Shallow Ice-Sheet model to simulate thickness and elevation evolution under artificial, transient changes in surface mass balance (SMB), similar to Practical 5. Then, we will apply an adjustable isotope–elevation sensitivity parameter (‰ per 100 m) based on the modeled thickness to translate modeled elevation changes into isotopic anomalies. Modeled results will be compared to the observed $\delta^{18}O$ records from the 6 sites.

Perturbing the model using real-world parameters

a) **Sloped bed:** Introduce a gentle retrograde slope to examine how topography amplifies or suppresses elevation sensitivity.

b) **Temperature-dependent rheology:** Implement an advection-diffusion heat-flow model from class (used in Practical 7) to compute vertical temperature profiles and adjust the Glen flow rate factor $A(T)$, reflecting faster flow in warmer ice.

Back-up

In case the time scope of this project is too far back such that the assumptions made of SMB is inadequate, the same Masson-Delmotte, V et al. dataset also includes the present interglacial $\delta^{18}O$ time series of the same 6 Antarctica ice cores (30 ka to present). Hence, the same project can be conducted based on modern SMB and bed topography.